COMPUTER VISION

ASSIGNMENT - 4

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Task 2 : Unique People Counting

2.1 Manual Count

The video was reviewed manually to identify distinct people appearing throughout the clip.

Total Manual Count: 7 People

Each individual was counted once, regardless of how many frames they appeared in.

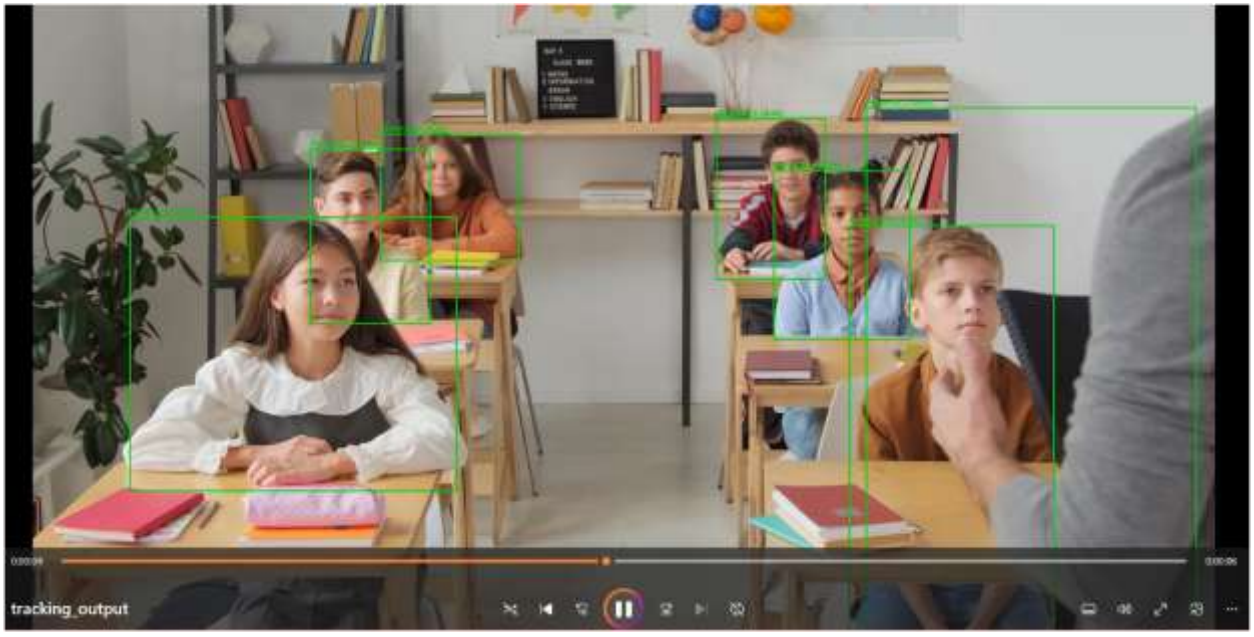


Figure 1: *Manual Count of Seven Distinct People in the Input Video for Comparison with the Tracking-Based Count.*

2.2 Automated Tracking Count

The tracking log (tracking_log.csv) was analyzed to extract all unique tracking IDs generated during video processing.

The following unique IDs were identified:

1, 2, 3, 4, 5, 6, 7, 8, 17, 19, 26, 28, 34

Total Unique Tracking IDs: 13

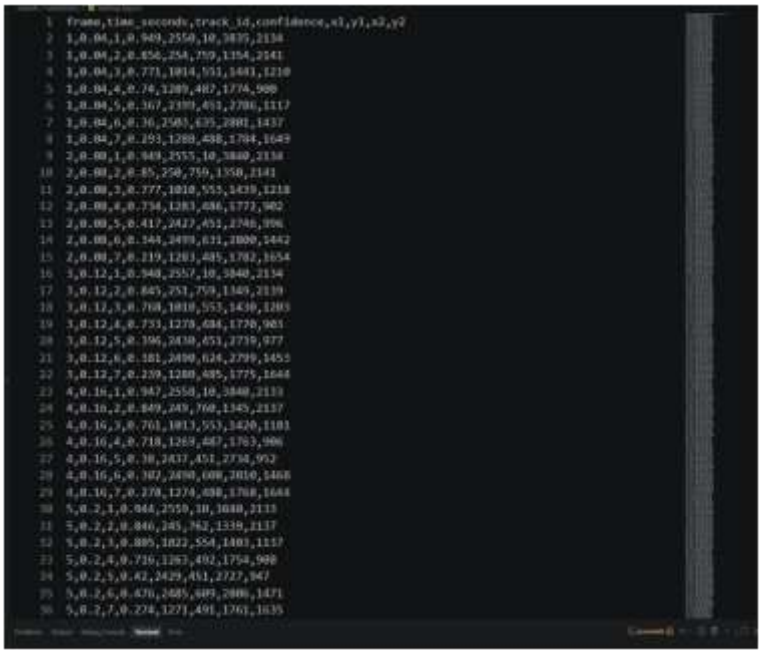


Figure 2: *Tracking Log (CSV) Showing Frame Numbers, Timestamps, Tracking IDs, Confidence Scores, and Bounding-Box Coordinates*.

2.3 Comparison

Counting Method	Number
Manual count	7
Unique tracking IDs	13
Difference	6

2.4 Explanation of the Difference

The tracking-based count is higher than the manual count because the tracking algorithm does not always maintain the same ID for a person throughout the video.

When a person becomes temporarily occluded or is not detected in consecutive frames, the tracker may lose the existing track. When that person is detected again, a new tracking ID may be assigned.

As a result, a single person can receive multiple tracking IDs during the same video. Therefore, the number of unique tracking IDs does not necessarily represent the actual number of distinct people.

2.5 Terminal Verification:

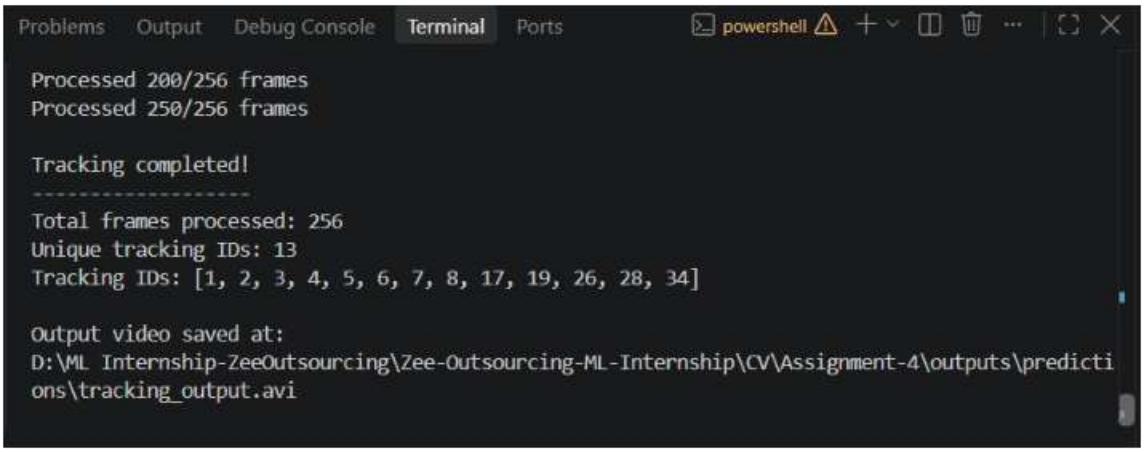


Figure 3: *Terminal Output Showing Successful Tracking Completion and 13 Unique Tracking IDs Across 256 Video Frames.*

Task 3 : Identifying a Tracking Identity Change

During the analysis of the tracking output, an identity change was observed involving the boy wearing a brown shirt.

Detail	Observation
Person	Boy wearing a brown shirt
Initial Tracking ID	8
New Tracking ID	19
Approximate Time	5 seconds
Tracking Issue	Track Fragmentation

Initially, the boy was assigned tracking ID 8. Around the 5-second mark, the same person was assigned a new tracking ID, 19.



Figure 1*: Brown-Shirt Boy Assigned Tracking ID 8 (Initial Tracking).*



Figure 2: *The Same Brown-Shirt Boy Assigned Tracking ID 19 (Later Tracking), Demonstrating Track Fragmentation.*

The person remained the same, but the tracking identity changed.

3.2 Explanation

This issue is known as **track fragmentation**. It occurs when a tracking algorithm loses an existing track and creates a new track for the same object.

Possible reasons include:

- Temporary occlusion by another person or object.
- Missed detections in consecutive frames.
- Changes in the position or size of the bounding box.
- Failure to associate a new detection with an existing track.

In this case, the tracker failed to maintain the original identity of the brown-shirt boy and assigned a new ID when tracking continued.

3.3 Track Fragmentation vs. ID Switch

Track fragmentation and ID switches are two different tracking problems.

- **Track Fragmentation:** The same person receives different IDs at different points in the video.
- **ID Switch:** An existing tracking ID is incorrectly assigned to a different person.

The observed change from ID 8 to ID 19 is an example of **track fragmentation**, rather than a confirmed ID switch.

3.4 Result

The tracking output demonstrated that maintaining consistent identities across video frames can be challenging, especially when people are temporarily difficult to detect. This observation highlights one of the limitations of multi-object tracking systems.